

1. Project Title: IoT Lab Inventory Management

2. Introduction

The objective of this project is to develop a robust and user-friendly website for managing the inventory of an IoT laboratory. The website aims to streamline the process of issuing components to students, tracking inventory in real time, managing running and completed projects, and facilitating communication between students and administrators.

3. Features of the Application

3.1 Inventory Management

- **Six Main Categories:**
 1. Sensors
 2. Microcontroller Boards
 3. Actuators
 4. Electronic Components
 5. Display Devices
 6. Miscellaneous
- **Subcategories** within each category:
 - Example:
 - **Sensors** → DHT11, LM35, PIR, IR, etc.
 - **Microcontroller Boards** → Arduino Uno, ESP32, NodeMCU, Raspberry Pi, etc.
- **Live Inventory Tracking:**
 - Real-time view of available stock in each subcategory.
 - Automatically updates on issue or return of items.

3.2 Component Issuance Form

- **Fields:**

- Student Name
- Roll Number
- Contact Number
- Date of Issue
- List of Requested Components
- Selection of components from available inventory.

3.3 Student Registration

- Only college email ID allowed.
- Upload recent passport-size photo during account creation.
- User login with OTP or secure password.

3.4 Admin Features

- Dashboard with:
 - Current inventory summary
 - Issued components summary
 - List of all registered students
- **Threshold Alerts:**
 - Admin receives **push notification** if any component quantity falls below minimum threshold.
- **Generate Requirement PDF:**
 - Automatically compile low-stock components into a formatted PDF for procurement or approval.
- **Send Messages / Announcements** to all registered users.

3.5 Project Tracking

- Add and track project status:
 - Project Title
 - Students involved
 - Status: **Running / Completed**

- Maintain a list of all:
 - **Student Projects**
 - **Consultancy Projects** with industry/government.

3.6 Events and Notification

- Admin can **broadcast event announcements** (e.g., workshops, competitions).
 - All registered students receive push notifications.
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4. Technology Stack

Feature	Technology
Frontend	Flutter / React Native
Backend	Firebase / Node.js with MongoDB
Authentication	Firebase Auth with college email domain validation
Image Upload	Firebase Storage / Cloudinary
Notification	Firebase Cloud Messaging (FCM)
PDF Generation	Flutter PDF / Node.js PDFKit
Database	Firebase Realtime Database / MongoDB Atlas

5. Security and Privacy

- Email ID validation ensures only college students register.
 - Student photos and data stored securely with encrypted access.
 - Admin access protected with authentication and authorization.
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6. Future Enhancements

- QR code-based component tracking.
- Auto-reminders for overdue returns.

- Component return form.
 - Downloadable reports for faculty/management.
 - Integration with college ERP or lab attendance system.
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7. Benefits

- Transparent and efficient component tracking.
 - Reduces human error and misuse of lab resources.
 - Improves communication and engagement with students.
 - Simplifies project and inventory reporting for administrators.
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8. Screens to be Developed (UI Overview)

- 1. Login / Registration Screen**
- 2. Home Dashboard (Student / Admin View)**
- 3. Component Categories & Subcategories**
- 4. Issue Component Form**
- 5. My Issued Components / History**
- 6. Admin Panel**
 - Inventory Management
 - Low Stock Alerts
 - Project Tracker
 - Student Database
 - Broadcast Message
- 7. Project Status Viewer**
- 8. Event Notification Section**
- 9. Requirement PDF Generator**